

Mason–Stothers via the Pandrosion Field: The Polynomial abc Theorem from Q -Tower Vanishing

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Abstract

We give a proof of the Mason–Stothers theorem (the polynomial abc inequality) using the Pandrosion field $F_P(z) = P'/P = \sum m_j/(z - \alpha_j)$ and the Q -tower vanishing from Paper 54. The Wronskian $W(a, b) = a'b - ab'$ decomposes as $W/ab = F_a - F_b$ (difference of Pandrosion fields). At each multiple root α of a (multiplicity m), the Q -tower vanishing $Q_a(\alpha, \alpha) = a'(\alpha) = 0$ forces $(z - \alpha)^{m-1} \mid W$; the same holds for multiple roots of b and $c = a + b$. Counting these forced zeros against $\deg W \leq \deg a + \deg b - 1$ yields $\max(\deg a, \deg b, \deg c) \leq N_0(abc) - 1$.

1 The Mason–Stothers theorem

Theorem 1.1 (Mason [1], Stothers [2]). *Let $a, b, c \in \mathbb{C}[z]$ be coprime polynomials (not all constant) satisfying $a + b + c = 0$. Then:*

$$\max(\deg a, \deg b, \deg c) \leq N_0(abc) - 1, \quad (1)$$

where $N_0(f)$ denotes the number of distinct zeros of f .

This is the polynomial analogue of the abc conjecture for integers.

2 The Pandrosion field

Definition 2.1. For $P(z) = \prod (z - \alpha_j)^{m_j}$ (with distinct roots $\alpha_1, \dots, \alpha_r$), the *Pandrosion field* is:

$$F_P(z) = \frac{P'(z)}{P(z)} = \sum_{j=1}^r \frac{m_j}{z - \alpha_j}. \quad (2)$$

It has exactly $N_0(P)$ simple poles at the distinct roots of P .

Proposition 2.2 (Wronskian as field difference). *For nonzero coprime polynomials a and b :*

$$\frac{W(a, b)}{a \cdot b} = F_a(z) - F_b(z), \quad (3)$$

where $W(a, b) = a'b - ab'$ is the Wronskian. Note that $W(a, b) \neq 0$ and $\deg W \leq \deg a + \deg b - 1$.

Proof. $W(a, b)/ab = a'/a - b'/b = F_a - F_b$. □

3 Proof of Mason–Stothers

Proof of Theorem 1.1 via Pandrosion. Write $c = -(a+b)$, with $\gcd(a, b) = \gcd(a, c) = \gcd(b, c) = 1$, and WLOG $\deg c = \max(\deg a, \deg b, \deg c)$.

Step 1: Wronskian. Set $W = W(a, b) = a'b - ab'$. Since a and b are coprime and not both constant, $W \neq 0$. Moreover:

$$\deg W \leq \deg a + \deg b - 1. \quad (\star)$$

Step 2: Zeros from multiple roots of a . Let α be a root of a with multiplicity $m \geq 2$. By the *Q-tower vanishing* (Paper 54):

$$Q_a(\alpha, \alpha) = a'(\alpha) = 0 \quad \text{and} \quad (z - \alpha)^{m-1} \mid a'.$$

Since $(z - \alpha)^m \mid a$ and $b(\alpha) \neq 0$ (coprimality):

$$W(\alpha) = \underbrace{a'(\alpha)}_{=0} \cdot b(\alpha) - \underbrace{a(\alpha)}_{=0} \cdot b'(\alpha) = 0.$$

More precisely, $(z - \alpha)^{m-1}$ divides W , contributing $m - 1$ zeros.

Step 3: Zeros from multiple roots of b . Identically, each root β of b with multiplicity $n \geq 2$ contributes $n - 1$ zeros of W .

Step 4: Zeros from multiple roots of c . Since $a + b = -c$: $W(a, b) = a'(-c) - a(-c)' = -(a'c - ac') = -W(a, c)$. So zeros of $W(a, c)$ are zeros of $W(a, b)$. Each root γ of c with multiplicity $p \geq 2$ contributes $p - 1$ zeros.

Step 5: Zero count. Writing $a = \prod (z - \alpha_j)^{m_j}$, $b = \prod (z - \beta_k)^{n_k}$, $c = \prod (z - \gamma_\ell)^{p_\ell}$:

$$\deg W \geq \sum (m_j - 1) + \sum (n_k - 1) + \sum (p_\ell - 1). \quad (4)$$

Since $\sum m_j = \deg a$, $\sum n_k = \deg b$, $\sum p_\ell = \deg c$:

$$\text{RHS} = (\deg a - N_0(a)) + (\deg b - N_0(b)) + (\deg c - N_0(c)). \quad (5)$$

Step 6: Combining with $(\star)$.

$$\begin{aligned} \deg a + \deg b - 1 &\geq \deg a - N_0(a) + \deg b - N_0(b) + \deg c - N_0(c), \\ N_0(a) + N_0(b) + N_0(c) - 1 &\geq \deg c, \\ \deg c &\leq N_0(abc) - 1. \end{aligned} \quad \square$$

4 The Pandrosion perspective

Remark 4.1 (Role of the *Q-tower*). The key step in the proof is Step 2: the *Q-tower vanishing* at multiple roots. This is precisely the Pandrosion characterization of multiplicity from Paper 54:

Multiplicity	<i>Q-tower</i>
α simple ($m = 1$)	$Q(\alpha, \alpha) = P'(\alpha) \neq 0$
α double ($m = 2$)	$Q(\alpha, \alpha) = 0, Q'(\alpha, \alpha) \neq 0$
α of mult. m	$Q^{(k)}(\alpha, \alpha) = 0$ for $k < m - 1$

Each vanishing level contributes one zero to the Wronskian, and the deficit $m - 1$ between multiplicity and distinctness measures the “hidden structure” detected by the Pandrosion field.

Remark 4.2 (Connection to the number-theoretic abc). Mason–Stothers is to abc what $\mathbb{C}[z]$ is to $\mathbb{Z}$: the Pandrosion field $F_P = P'/P$ plays the role of the “arithmetic derivative,” and the Wronskian $W(a, b) = ab \cdot (F_a - F_b)$ captures the interaction between the two “ abc -terms.” The difficulty in passing from polynomials to integers lies in the absence of a global derivative in $\mathbb{Z}$.

5 Numerical verification

a	b	max deg	$N_0(abc)$	Bound	MS
z^2	$1 - z$	2	4	3	✓
z^3	$2 - 3z$	3	4	3	✓ (tight!)
$z^4 + 1$	$-z^2$	4	9	8	✓
$(z-1)^2(z+2)$	$z + 3$	3	6	5	✓

In the second example, $c = z^3 - 3z + 2 = (z - 1)^2(z + 2)$ has a double root; the Wronskian factors as $W = (z - 1)(2z^2 + 11z + 11)$, confirming the forced zero at $z = 1$ from Q -tower vanishing.

References

- [1] R. C. Mason, *Diophantine Equations over Function Fields*, London Math. Soc. Lecture Notes **96**, Cambridge, 1984.
- [2] W. W. Stothers, *Polynomial identities and Hauptmoduln*, Quart. J. Math. **32** (1981), 349–370.
- [3] S. Lang, *Algebra*, 3rd ed., Springer, 2002, Ch. IV.7.
- [4] I. Besevic, *Casas-Alvero for prime degree: the Q -tower approach*, Pandrosion Dynamics **54** (2026).